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29 APRIL 1966
SUPERSEDING
MIL-S-17346A
11 OCTOBER 1954

MILITARY SPECIFICATION

STONE, COMMUTATOR BURNISHING (FLEXIBLE ABRASIVE)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers flexible abrasive commutator burnishing stones of the aluminum oxide type having a resilient bond.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-P-3816 — Abrasives and Abrasive Products for Shipment and Storage, Packaging and Packing of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer).

3. REQUIREMENTS

3.1 **Material.** The commutator burnishing stones shall be made of crystalline aluminum oxide abrasive materials of a quality necessary to comply with the requirements specified herein.

3.2 **Design.** The stones shall be designed to be flexible, nondusting, nonloading and nonconductive and shall be suitable for producing a burnished surface on polished metal.

3.3 **Construction.** The abrasive grains used in construction of the stones shall be screened through a 100 mesh (minimum) screen. The grains shall be bonded with a flexible bond of synthetic rubber.

3.3.1 *Hardness.* Stones shall have a durometer hardness of not more than 95 (A scale) ASTM when tested as specified in 4.4.1.

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3.3.2 *Flexibility.* Stones shall be capable of being bent around a mandrel diameter ten times the smallest dimension of the stone without breaking or cracking (see 4.4.2).

3.3.3 *Dielectric strength.* Stones shall show a resistance of at least 10 megohms when tested as specified in 4.4.3.

3.4 **Size and dimensions.** The stones shall be of the following dimensions for size specified (see 6.2).

TABLE I. *Size and dimensions.*

Size	Dimensions in inches			Tolerance (Plus or minus) Percent
	Thickness	Width	Length	
1	1/8	1	5	10
2	3/16	1/2	5	10
3	1/4	3/4	5	10
4	1/4	1	5	10
5	1	2	5	10
6	1	4	5	10

3.5 **Identification.** Each stone shall be legibly and permanently marked with the manufacturer's identification.

3.6 **Workmanship.** Workmanship shall be of the quality prevailing among manufacturers normally producing burnishing stones of sizes, dimensions and materials listed herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Sampling.**

4.2.1 *Lot.* All stones of the same size presented for inspection at one time shall be considered a lot for purposes of sampling.

4.2.2 *Sampling for quality conformance inspection.* A random sample shall be selected from each lot in conformance with MIL–STD–105 at Inspection Level II.

4.3 **Quality conformance inspection.** Quality conformance inspection shall be applied to each sample selected in accordance with 4.2.2. Quality conformance

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inspection shall be applied to each sample selected in accordance with 4.4.2. Quality conformance inspection shall consist of the following:

(a) Examination (see 4.3.1).

(b) Tests (see 4.3.2 and 4.4).

(c) Examination of preservation, packaging, packing and storage (see 4.5).

4.3.1 *Examination.* Each of the sample stones shall be examined for design, size and dimensions and workmanship to determine conformance with this specification. Examination shall be conducted in accordance with table II. The acceptable quality level (AQL) shall be the following: Major —2.5 percent defective and minor—4.0 percent defective. Nonconforming stones shall not be offered for delivery and if the nonconforming stones in any one sample exceed the acceptance number for that sample, this shall be cause for rejection of the entire lot represented by the sample.

TABLE II. *Classification of defects.*

Categories	Defects
Major:	
101	Size and dimensions not as specified.
Minor:	
201	Identification, marking illegible.
202	Evidence of poor workmanship, physical shape irregular.

4.3.2 *Tests.* Each sample selected in accordance with 4.2.2 shall be tested as specified in 4.4 to determine compliance with the requirements of this specification. The AQL shall be 2.5 percent defective. Nonconforming stones shall not be offered for delivery and if the number of defective stones exceeds the acceptable number for the samples tested, this shall be cause for rejection of the lot represented by the sample.

4.4 Test procedures.

4.4.1 *Hardness.* Each sample stone shall be tested in accordance with method 3041 of FED-STD-601 and the ASTM hardness number shall not exceed 95 on the A scale.

4.4.2 *Flexibility.* Each sample stone shall be tested for flexibility by bending in such a way that the shortest dimension is normal to the testing mandrel, the second shortest dimension is parallel to the axis of the mandrel and the longest dimension is bent around the mandrel. Mandrel diameter shall not be more than ten times the smallest dimension of stone. Evidence of cracking or breaking shall be cause for rejection.

4.4.3 *Dielectric strength.* Stones shall be tested with a potential of 6 to 10 volts using two 0.050 to .065 inch diameter test prods inserted at least 1/4 inch into stone and not less than 1/8 inch or more than 1/2 inch apart. A resistance of less than 10 megohms shall be cause for rejection.

4.4.4 *Burnishing action.* The commutator burnishing stone shall be pressed against the surface of a commutator or electrical contact ring rotating at a surface speed from 720 to 760 surface feet per minute and which has previously been conditioned with commutator dressing stones. Burnishing action shall be noted. Defects visible to the unaided eye such as scratches on the burnished surface and retention of metallic particles shall be cause for rejection. A film of microscopic metallic particles on the surface of the stone with no separate particles visible shall not be cause for rejection.

4.5 Inspection of preservation, packaging, packing and marking for shipment and

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storage. Preservation, packaging, packing and marking for shipment and storage shall be inspected to determine conformance with the documents referenced in section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be in accordance with the requirements of MIL-P-3816 for stones, commutator. Levels of preservation and packaging and level of packing shall be as specified (see 6.2).

5.2 Marking. Unless otherwise specified (see 6.2), marking shall conform to MIL-STD-129.

6. NOTES

6.1 Intended use. Burnishing stones as furnished under this specification are flexible abrasive stones intended for removing high resistance oxide films and for final polishing and burnishing of electric motor commutator and collector rings which have been conditioned with commutator dressing stones.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.

(b) Size required (see 3.4).

(c) Level of preservation and packaging and level of packing required (see 5.1).

(d) Special marking, if required (see 5.2).

Custodians:

Army—MR

Navy—SH

Air Force—69

Review activities:

Army—MR

Navy—SH

Air Force—69, 85

User activities:

Army—MU, EL

Navy—MC

Preparing activity:

Navy—SH

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